

Laporan Hasil Testing

Report generated on 27-Apr-2026 at 11:01:51 by [pytest-html](#) v4.2.0

Environment

Python	3.10.0
Platform	Windows-10-10.0.26200-SP0
Packages	- pytest: 8.3.5 - pluggy: 1.6.0
Plugins	- anyio: 4.13.0 - asyncio: 0.25.3 - html: 4.2.0 - metadata: 3.1.1

Summary

199 tests took 00:00:59.

(Un)check the boxes to filter the results.

☒ 0 Failed, ☒ 199 Passed, ☒ 0 Skipped, ☒ 0 Expected failures, ☒ 0 Unexpected passes, ☒ 0 Errors, ☒ 0 Reruns ☒ 0 Retried,

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Result 	Test	Deskripsi	Duration	Links
Passed	tests/chatPipeline_test.py::test_process_query_returns_clarification_when_query_is_ambiguous	-	8 ms	
Passed	tests/chatPipeline_test.py::test_process_query_returns_security_message_when_sql_validation_fails	-	4 ms	
Passed	tests/chatPipeline_test.py::test_process_query_success_without_visualization	-	5 ms	
Passed	tests/chatPipeline_test.py::test_process_query_success_with_visualization	-	5 ms	
Passed	tests/chatPipeline_test.py::test_process_query_falls_back_when_analysis_rate_limited	-	5 ms	
Passed	tests/chatPipeline_test.py::test_process_query_propagates_timeout_http_exception	-	5 ms	
Passed	tests/chatPipeline_test.py::test_process_query_returns_fallback_message_when_llm_unavailable	-	4 ms	
Passed	tests/chatStreaming_test.py::test_handle_chat_returns_stream_with_message_words	-	2 ms	
Passed	tests/chatStreaming_test.py::test_handle_clarification_streams_message_and_keeps_metadata_non_stream	-	2 ms	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_admin_can_access_chatbot_module	Role admin boleh akses module chatbot.	00:00:02	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_non_admin_cannot_access_chatbot_module[hrd]	Role non-admin harus ditolak ketika akses module chatbot.	00:00:03	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_non_admin_cannot_access_chatbot_module[kepala_divisi]	Role non-admin harus ditolak ketika akses module chatbot.	00:00:03	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_non_admin_cannot_access_chatbot_module[karyawan]	Role non-admin harus ditolak ketika akses module chatbot.	00:00:03	
Passed	tests/chatbotManagement_test.py::TestChatbotRBAC::test_chatbot_module_requires_token	Tanpa token, endpoint chatbot harus return 401.	20 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_hrd_success	POST chatbot HRD dengan semua field lengkap harus return 201.	880 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_karyawan_success	POST chatbot Karyawan harus return 201.	877 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_without_addon_prompt	addon_prompt bersifat opsional — boleh null.	934 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_same_otoritas_deactivates_previous_active	Create chatbot aktif baru dengan otoritas sama harus nonaktifkan chatbot aktif lama.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_duplicate_nama_returns_409	Nama chatbot yang sudah ada harus return 409 Conflict.	993 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_duplicate_nama_case_insensitive	Duplikat nama tidak peka huruf besar/kecil.	965 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_missing_nama_returns_422	nama_chatbot wajib — jika tidak ada harus return 422.	937 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_missing_otoritas_returns_422	otoritas wajib — jika tidak ada harus return 422.	802 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_invalid_otoritas_returns_422	otoritas hanya boleh 'HRD' atau 'Karyawan'.	958 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_empty_nama_returns_422	nama_chatbot tidak boleh string kosong.	843 ms	
Passed	tests/chatbotManagement_test.py::TestCreateChatbot::test_create_nama_stripped_whitespace	nama_chatbot dengan spasi di ujung harus di-strip.	902 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_empty	List chatbot saat DB kosong harus return data= [] dan total=0.	837 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_returns_all_active	Semua chatbot aktif harus muncul di list.	00:00:01	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_inactive_included	Chatbot dengan is_active=False tetap muncul di list default.	948 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_pagination_page_size	page_size membatasi jumlah item per halaman.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_pagination_page_2_different_items	Halaman 2 harus return item yang berbeda dari halaman 1.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_invalid_page_returns_422	page < 1 harus return 422.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_hrd	Filter otoritas=HRD hanya return chatbot HRD.	967 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_karyawan	Filter otoritas=Karyawan hanya return chatbot Karyawan.	922 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_filter_invalid_otoritas_returns_422	Filter dengan nilai otoritas tidak valid harus return 422.	963 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_search_by_nama	Search berdasarkan nama_chatbot (partial match).	00:00:01	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_search_by_addon_prompt	Search juga mencakup konten addon_prompt.	892 ms	
Passed	tests/chatbotManagement_test.py::TestListChatbot::test_list_search_no_result	Search yang tidak cocok harus return data kosong.	980 ms	
Passed	tests/chatbotManagement_test.py::TestGetChatbotById::test_get_by_id_success	GET /{id} harus return detail chatbot yang benar.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestGetChatbotById::test_get_by_id_not_found	GET dengan ID yang tidak ada harus return 404.	941 ms	
Passed	tests/chatbotManagement_test.py::TestGetChatbotById::test_get_inactive_returns_200	Chatbot nonaktif tetap bisa diakses via GET /{id}.	998 ms	
Passed	tests/chatbotManagement_test.py::TestGetChatbotById::test_get_invalid_id_returns_422	ID bukan integer harus return 422.	981 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_nama_only	Update hanya nama_chatbot — field lain tidak berubah.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_otoritas_only	Update hanya otoritas — nama tidak berubah.	888 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_addon_prompt	Update addon_prompt dengan nilai baru.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_addon_prompt_to_null	addon_prompt bisa di-set ke null.	982 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_all_fields	Update semua field sekaligus.	932 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_is_active_to_false	is_active bisa di-update melalui PATCH.	975 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_activate_chatbot_deactivates_other_same_otoritas	Saat aktivasi chatbot, chatbot aktif lain pada otoritas sama harus dinonaktifkan.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_otoritas_active_chatbot_deactivates_active_target_otoritas	Pindah otoritas chatbot aktif harus nonaktifkan chatbot aktif lain di otoritas tujuan.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_nama_to_existing_returns_409	Ganti nama ke nama chatbot lain yang sudah ada harus return 409.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_same_nama_not_conflict	Update ke nama yang sama (milik dirinya sendiri) tidak boleh 409.	00:00:01	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_not_found_returns_404	Update chatbot yang tidak ada harus return 404.	967 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_invalid_otoritas_returns_422	otoritas tidak valid saat update harus return 422.	864 ms	
Passed	tests/chatbotManagement_test.py::TestUpdateChatbot::test_update_empty_nama_returns_422	nama_chatbot tidak boleh string kosong saat update.	935 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_returns_200	Soft delete harus return 200 dengan pesan sukses.	995 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_sets_is_active_false	Setelah soft delete, is_active di DB harus False.	877 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_still_accessible_via_get	Setelah soft delete, GET /{id} tetap bisa diakses dengan status nonaktif.	997 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_still_in_list	Setelah soft delete, chatbot tetap muncul di GET / dengan status nonaktif.	754 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_not_found_returns_404	Soft delete chatbot yang tidak ada harus return 404.	194 ms	
Passed	tests/chatbotManagement_test.py::TestSoftDeleteChatbot::test_soft_delete_twice_returns_200	Soft delete dua kali bersifat idempotent dan tetap return 200.	224 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_returns_200	Hard delete harus return 200 dengan pesan sukses.	197 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_removes_from_db	Setelah hard delete, record benar-benar hilang dari DB.	195 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_not_found_returns_404	Hard delete chatbot yang tidak ada harus return 404.	188 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_hard_delete_nama_can_be_reused	Setelah hard delete, nama yang sama bisa dipakai lagi.	201 ms	
Passed	tests/chatbotManagement_test.py::TestHardDeleteChatbot::test_default_delete_is_soft	Tanpa ?hard=true, default adalah soft delete — record masih ada di DB.	190 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_detects_ambiguity	Test LLM-based detection mendeteksi ambiguitas.	9 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_clear_query	Test LLM menganggap query yang jelas sebagai tidak ambiguous.	8 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_json_in_markdown_fence	Test parser tetap menerima JSON di dalam markdown code fence.	10 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_tie_breaking_rule	Test tie-breaking rule: score 0.55-0.65 treated as NOT ambiguous.	8 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_api_error_fallback	Test fallback behavior ketika LLM API error.	8 ms	
Passed	tests/clarificationMechanism_test.py::TestAmbiguityDetectorLLM::test_llm_invalid_json_fallback	Test fallback ketika LLM return invalid JSON.	10 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationQuestionGenerator::test_generate_question_from_llm	Test generate pertanyaan dari LLM.	8 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationQuestionGenerator::test_generate_question_llm_error_fallback	Test fallback ke template ketika LLM error.	8 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationService::test_process_direct_answer_no_ambiguity	Test direct answer ketika query tidak ambiguous.	21 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationService::test_process_clarification_needed	Test clarification diperlukan ketika query ambiguous.	21 ms	
Passed	tests/clarificationMechanism_test.py::TestClarificationService::test_max_clarification_limit	Test max clarification limit per session.	2 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_create_session_context	Test create new session context.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_add_clarification_to_history	Test add clarification to session history.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_scope_preference_storage	Test store scope preference dari clarification answer.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_preference_persistence_across_queries	Test scope preference persist across multiple queries.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestSessionContextManager::test_session_ttl_cleanup	Test session TTL cleanup mechanism.	0 ms	
Passed	tests/clarificationMechanism_test.py::TestScenarios::test_scenario_llm_ambiguous_query	Skenario: LLM mendeteksi ambiguitas dan generate pertanyaan.	31 ms	
Passed	tests/clarificationMechanism_test.py::TestScenarios::test_scenario_no_ambiguity	Skenario: LLM menganggap query sudah jelas.	9 ms	
Passed	tests/clarificationMechanism_test.py::TestScenarios::test_scenario_tie_breaking	Skenario: Score borderline (0.55-0.65) apply tie-breaking.	9 ms	
Passed	tests/clarificationMechanism_test.py::TestScenarios::test_scenario_llm_unavailable	Skenario: LLM tidak tersedia, fallback to safe default.	16 ms	
Passed	tests/kpiGroupCRUD_test.py::test_create_kpi_group_master_with_explicit_sheet_info	Create KPI Group master dengan sheet_id dan nama_grup yang diberikan.	19 ms	
Passed	tests/kpiGroupCRUD_test.py::test_create_kpi_group_auto_fetch_sheet_info	Create KPI Group auto-fetch sheet_id dan nama dari Google Sheets jika tidak diberikan.	4 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/kpiGroupCRUD_test.py::test_create_kpi_group_tracker	Create KPI Group dengan group_type=tracker.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_all	List semua KPI Groups dengan pagination.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_with_filter	List KPI Groups dengan filter group_type.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_with_search	List KPI Groups dengan search keyword.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_list_kpi_groups_pagination	List KPI Groups dengan pagination calculation.	5 ms	
Passed	tests/kpiGroupCRUD_test.py::test_get_kpi_group_by_id	Get satu KPI Group dengan all related records.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_get_kpi_group_not_found	Get KPI Group dengan ID yang tidak ada.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_get_kpi_group_master_with_records	Get master KPI Group dengan all master records.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_partial	Update beberapa field pada KPI Group.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_sheet_url_triggers_re_ingestion_master	Update sheet_url pada master group triggers KPI Master re-ingestion.	6 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_sheet_url_triggers_re_ingestion_tracker	Update sheet_url pada tracker group triggers Tracker re-ingestion.	21 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_master_requires_tahun_when_sheet_url_changes	Update sheet_url pada master group HARUS disertai tahun.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_update_kpi_group_not_found	Update KPI Group dengan ID yang tidak ada.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_delete_kpi_group	Delete KPI Group by ID.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_delete_kpi_group_commits_transaction	Delete operation commits transaction.	4 ms	
Passed	tests/kpiGroupCRUD_test.py::test_build_response_excludes_records_for_list	_build_response without include_records=True returns empty record lists.	3 ms	
Passed	tests/kpiGroupCRUD_test.py::test_build_response_includes_master_records	_build_response includes master_records when include_records=True.	20 ms	
Passed	tests/kpiGroupCRUD_test.py::test_build_response_fields_match_orm	_build_response correctly maps all ORM fields to response.	2 ms	
Passed	tests/kpiGroupCRUD_test.py::test_create_list_get_delete_flow	Complete CRUD flow: create → list → get → delete.	5 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingest_kpi_master_success	Successful KPI Master ingestion.	6 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingest_kpi_master_partial_success	KPI Master ingestion with some errors.	7 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingest_kpi_master_no_valid_records	Ingestion fails when no valid records found after parsing.	6 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingest_kpi_master_sheet_fetch_error	Ingestion fails when Google Sheets fetch fails.	14 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingest_kpi_master_unexpected_error_updates_log	Unexpected error during ingestion updates log to failed.	18 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingest_creates_kpi_group_if_not_exists	Ingestion creates KPIGroup if sheet_id not seen before.	7 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingest_upserts_existing_kpi_group	Re-ingesting same sheet updates existing KPIGroup.	6 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingestion_log_created_before_processing	IngestionLog created with status='running' before processing.	6 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingestion_log_updated_on_completion	IngestionLog updated with final status and counts.	7 ms	
Passed	tests/kpiMasterIngestion_test.py::test_ingest_injects_group_id_into_records	Each record receives group_id before upsert.	16 ms	
Passed	tests/kpiTrackerBatch_test.py::test_batch_returns_results_for_each_url	Each URL in the request gets its own UrlIngestionResult in the response.	11 ms	
Passed	tests/kpiTrackerBatch_test.py::test_batch_empty_url_list_rejected	Request with empty sheet_urls list should return 422.	5 ms	
Passed	tests/kpiTrackerBatch_test.py::test_batch_too_many_urls_rejected	Request with more than 20 URLs should return 422.	28 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/kpiTrackerBatch_test.py::test_batch_one_url_fails_others_still_succeed	A single failed URL should appear as failed=1, not abort the whole batch.	7 ms	
Passed	tests/promptTemplate_test.py::test_build_analysis_prompt_supports_datetime_values	-	0 ms	
Passed	tests/scheduler_test.py::TestSchedulerRepository::test_get_config_returns_none_when_empty	get_config mengembalikan None jika belum ada config di DB.	3 ms	
Passed	tests/scheduler_test.py::TestSchedulerRepository::test_create_config_add_commit_refresh	create_config harus add + commit + refresh object SchedulerConfigORM.	4 ms	
Passed	tests/scheduler_test.py::TestSchedulerRepository::test_create_config_rollback_and_raise_http_500	Jika commit gagal, repository wajib rollback lalu melempar HTTP 500.	3 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_build_trigger_returns_cron_trigger	_build_trigger harus menghasilkan CronTrigger.	0 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_build_trigger_fires_on_correct_day_and_hour	CronTrigger harus fire pada hari dan jam yang benar tiap bulan.	1 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_auto_pause_if_december_pauses_in_december	_auto_pause_if_december harus remove job dan set is_enabled=False di bulan Desember.	5 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_auto_pause_if_december_skips_non_december	_auto_pause_if_december tidak boleh pause di bulan selain Desember.	1 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_register_job_enabled_adds_cron_job	register_job harus menambah CronTrigger job saat config enabled.	2 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_register_job_disabled_does_not_add_job	register_job tidak boleh add job jika scheduler dinonaktifkan.	1 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_register_job_replaces_existing_job	Jika job lama ada, register_job harus remove lalu add yang baru.	2 ms	
Passed	tests/scheduler_test.py::TestSchedulerService::test_get_next_run_time_returns_none_when_job_missing	get_next_run_time harus None ketika job scheduler belum ada.	1 ms	
Passed	tests/scheduler_test.py::TestSchedulerController::test_create_config_save_and_register_scheduler_job	create_config harus simpan config baru, register job, dan isi next_run_at.	4 ms	
Passed	tests/scheduler_test.py::TestSchedulerController::test_create_config_when_existing_returns_409	create_config harus menolak bila config scheduler sudah ada.	3 ms	
Passed	tests/scheduler_test.py::TestSchedulerController::test_trigger_now_runs_ingestion_job	trigger_now harus mengeksekusi _run_ingestion_job saat config tersedia.	3 ms	
Passed	tests/scheduler_test.py::TestSchedulerController::test_trigger_now_without_config_returns_404	trigger_now harus return 404 jika config scheduler belum dibuat.	2 ms	
Passed	tests/user_test.py::TestLogin::test_login_sukses	Login valid mengembalikan access token, refresh token, dan data user.	5 ms	
Passed	tests/user_test.py::TestLogin::test_login_password_salah	Login dengan password salah harus ditolak 401.	5 ms	
Passed	tests/user_test.py::TestLogin::test_login_user_tidak_ditemukan	Login dengan identifier yang tidak terdaftar harus return 401.	4 ms	
Passed	tests/user_test.py::TestLogin::test_login_akun_nonaktif	User nonaktif tidak boleh login dan harus mendapat 403.	4 ms	
Passed	tests/user_test.py::TestLogin::test_login_via_email	Login via email (bukan username) tetap valid dan mengembalikan 200.	4 ms	
Passed	tests/user_test.py::TestLogin::test_login_payload_kosong	Payload login kosong harus gagal validasi request (422).	3 ms	
Passed	tests/user_test.py::TestRefresh::test_refresh_sukses_rotation	Refresh token valid harus merotasi token dan menghasilkan token baru.	22 ms	
Passed	tests/user_test.py::TestRefresh::test_refresh_token_sudah_direvoke	Refresh token yang sudah direvoke harus ditolak 401.	5 ms	
Passed	tests/user_test.py::TestRefresh::test_refresh_token_kadaluarsa	Refresh token kadaluarsa harus ditolak 401.	4 ms	
Passed	tests/user_test.py::TestRefresh::test_refresh_token_palsu	Refresh token palsu dengan format JWT tidak valid harus ditolak 401.	4 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/user_test.py::TestRefresh::test_refresh_kirim_access_token	Access token dikirim ke /refresh → ditolak karena type mismatch.	3 ms	
Passed	tests/user_test.py::TestRefresh::test_refresh_user_nonaktif	Refresh token milik user nonaktif harus ditolak 403.	5 ms	
Passed	tests/user_test.py::TestRefresh::test_refresh_user_dihapus	Refresh token untuk user yang sudah tidak ada harus ditolak 401.	5 ms	
Passed	tests/user_test.py::TestRefresh::test_refresh_revoke_dipanggil_sekali	Pada refresh sukses, revoke_token harus dipanggil tepat satu kali.	5 ms	
Passed	tests/user_test.py::TestLogout::test_logout_sukses	Logout valid harus revoke refresh token dan mengembalikan 200.	6 ms	
Passed	tests/user_test.py::TestLogout::test_logout_token_palsu	Logout dengan refresh token palsu harus ditolak 401.	3 ms	
Passed	tests/user_test.py::TestLogout::test_logout_token_kadaluarsa	Logout dengan refresh token kadaluarsa harus ditolak 401.	3 ms	
Passed	tests/user_test.py::TestLogout::test_logout_idempotent	Logout dua kali tidak boleh error (revoke_token bersifat idempotent).	6 ms	
Passed	tests/user_test.py::TestGetMe::test_get_me_sukses	Semua role bisa akses /users/me (ALL_ROLES di middleware).	4 ms	
Passed	tests/user_test.py::TestGetMe::test_get_me_sebagai_hrd	HRD juga bisa akses /users/me.	4 ms	
Passed	tests/user_test.py::TestGetMe::test_get_me_sebagai_kepala_divisi	Kepala divisi juga bisa akses /users/me.	5 ms	
Passed	tests/user_test.py::TestGetMe::test_get_me_tanpa_token	Endpoint /users/me tanpa Authorization header harus return 401.	2 ms	
Passed	tests/user_test.py::TestGetMe::test_get_me_token_kadaluarsa	Endpoint /users/me dengan access token kadaluarsa harus return 401.	2 ms	
Passed	tests/user_test.py::TestGetMe::test_get_me_kirim_refresh_token	Refresh token tidak boleh lolos sebagai access token.	2 ms	
Passed	tests/user_test.py::TestChangePassword::test_change_password_sukses	User terautentikasi dapat mengganti password jika old password benar.	183 ms	
Passed	tests/user_test.py::TestChangePassword::test_change_password_lama_salah	Pergantian password harus gagal jika old password tidak cocok.	5 ms	
Passed	tests/user_test.py::TestChangePassword::test_change_password_sama	Password baru tidak boleh sama dengan password lama.	5 ms	
Passed	tests/user_test.py::TestChangePassword::test_change_password_validasi_kekuatan	Password baru tanpa angka → 422 dari Pydantic validator.	4 ms	
Passed	tests/user_test.py::TestChangePassword::test_change_password_tanpa_auth	Endpoint change-password tanpa token harus ditolak 401.	2 ms	
Passed	tests/user_test.py::TestCreateUser::test_create_user_sukses	Admin dapat membuat user baru ketika username/email belum terpakai.	194 ms	
Passed	tests/user_test.py::TestCreateUser::test_create_user_duplikat_username	Pembuatan user harus gagal 409 jika username sudah terdaftar.	15 ms	
Passed	tests/user_test.py::TestCreateUser::test_create_user_duplikat_email	Pembuatan user harus gagal 409 jika email sudah terdaftar.	5 ms	
Passed	tests/user_test.py::TestCreateUser::test_create_user_forbidden_sebagai_hrd	HRD tidak boleh membuat user baru (POST /users hanya admin).	3 ms	
Passed	tests/user_test.py::TestCreateUser::test_create_user_forbidden_sebagai_kepala_divisi	Kepala divisi tidak boleh membuat user baru.	3 ms	
Passed	tests/user_test.py::TestCreateUser::test_create_user_forbidden_sebagai_karyawan	Karyawan tidak boleh membuat user baru.	3 ms	
Passed	tests/user_test.py::TestCreateUser::test_create_user_password_lemah	Password tanpa huruf kapital → 422.	4 ms	
Passed	tests/user_test.py::TestGetAllUsers::test_get_all_users_sukses	Admin dapat melihat daftar user dengan metadata total yang benar.	5 ms	
Passed	tests/user_test.py::TestGetAllUsers::test_get_all_users_pagination	Parameter limit dan offset harus diteruskan ke repository dengan benar.	5 ms	

Result 	Test	Deskripsi	Duration	Links
Passed	tests/user_test.py::TestGetAllUsers::test_get_all_users_forbidden_sebagai_hrd	HRD tidak boleh akses list semua user.	3 ms	
Passed	tests/user_test.py::TestGetAllUsers::test_get_all_users_forbidden_sebagai_karyawan	Karyawan tidak boleh akses list semua user.	4 ms	
Passed	tests/user_test.py::TestGetUserById::test_get_user_by_id_sukses	Admin dapat mengambil detail user berdasarkan user_id yang valid.	5 ms	
Passed	tests/user_test.py::TestGetUserById::test_get_user_by_id_not_found	Pengambilan detail user harus return 404 jika user_id tidak ditemukan.	4 ms	
Passed	tests/user_test.py::TestGetUserById::test_get_user_by_id_forbidden_sebagai_karyawan	Karyawan tidak boleh akses GET /users/{id} selain /users/me. Middleware: GET ^/api/v1/users/{w-}+\$ → ALL_ROLES, tapi rule ini hanya berlaku untuk /users/me. Pastikan rule di middleware sudah memisahkan /me vs /{id}.	4 ms	
Passed	tests/user_test.py::TestGetUserById::test_get_user_by_id_forbidden_sebagai_hrd	HRD tidak boleh akses GET /users/{id}.	4 ms	
Passed	tests/user_test.py::TestUpdateUser::test_update_user_sukses	Admin dapat memperbarui data user jika target ada dan email tidak konflik.	8 ms	
Passed	tests/user_test.py::TestUpdateUser::test_update_user_email_duplikat	Update email harus ditolak 409 jika email baru sudah dipakai user lain.	5 ms	
Passed	tests/user_test.py::TestUpdateUser::test_update_user_not_found	Update user harus return 404 ketika user target tidak ditemukan.	4 ms	
Passed	tests/user_test.py::TestUpdateUser::test_update_user_forbidden_sebagai_karyawan	Karyawan tidak boleh PATCH /users/{id} milik orang lain.	23 ms	
Passed	tests/user_test.py::TestDeleteUser::test_delete_user_sukses	Admin dapat menghapus user lain dan menerima pesan sukses.	5 ms	
Passed	tests/user_test.py::TestDeleteUser::test_delete_diri_sendiri	Admin tidak dapat menghapus akunnnya sendiri.	4 ms	
Passed	tests/user_test.py::TestDeleteUser::test_delete_user_not_found	Delete user harus return 404 jika user target tidak ditemukan.	4 ms	
Passed	tests/user_test.py::TestDeleteUser::test_delete_user_forbidden_sebagai_hrd	HRD tidak boleh menghapus user.	3 ms	
Passed	tests/user_test.py::TestDeleteUser::test_delete_user_forbidden_sebagai_kepala_divisi	Kepala divisi tidak boleh menghapus user.	3 ms	
Passed	tests/user_test.py::TestDeleteUser::test_delete_user_forbidden_sebagai_karyawan	Karyawan tidak boleh menghapus user.	2 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_hash_dan_verify_password	hash_password menghasilkan hash bcrypt yang lolos verify_password untuk nilai benar.	536 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_create_access_token_berisi_field_wajib	Access token harus memuat sub, username, role, type bearer, dan expiry positif.	1 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_create_refresh_token_berisi_field_minimal	Refresh token hanya memuat field minimal tanpa role/username.	1 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_decode_access_token_tolak_refresh_token	decode_access_token harus menolak token bertipe refresh.	1 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_decode_refresh_token_tolak_access_token	decode_refresh_token harus menolak token bertipe access.	1 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_decode_token_kadaluarsa	decode_access_token harus melempar 401 untuk token yang sudah kadaluarsa.	1 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_authenticate_user_sukses	authenticate_user mengembalikan user saat identifier dan password valid.	24 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_rotate_tokens_revoke_lama_terbit_baru	rotate_tokens harus revoke token lama dan menerbitkan pasangan token baru.	9 ms	
Passed	tests/user_test.py::TestAuthServiceUnit::test_rotate_tokens_reuse_ditolak	rotate_tokens harus menolak refresh token yang terdeteksi reuse/revoked.	26 ms	